ReactJS

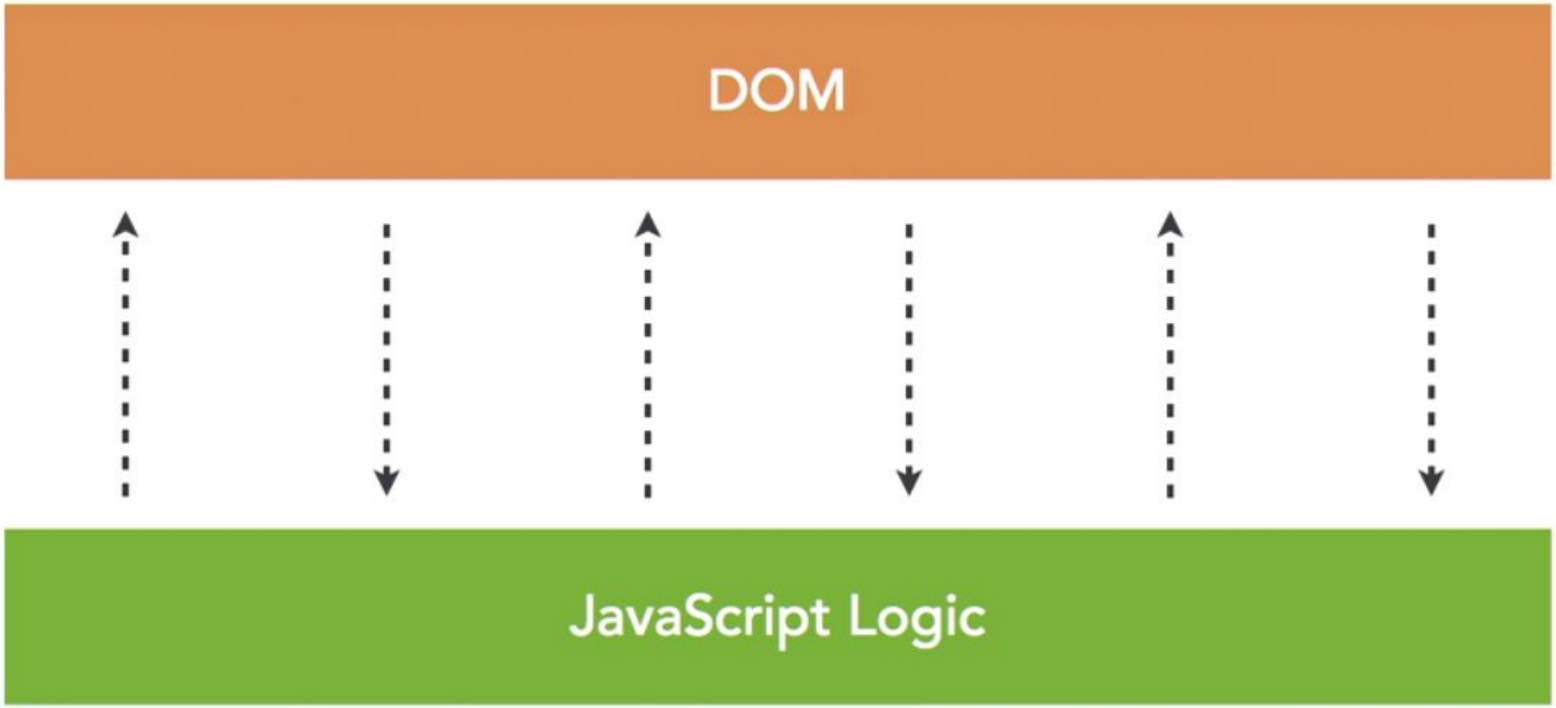
Introduction

- Une librairie pour créer des interfaces utilisateur.
- Creer par Facebook et Instagram (mars 2013)
- Suivi par React native pour les téléphones mobiles

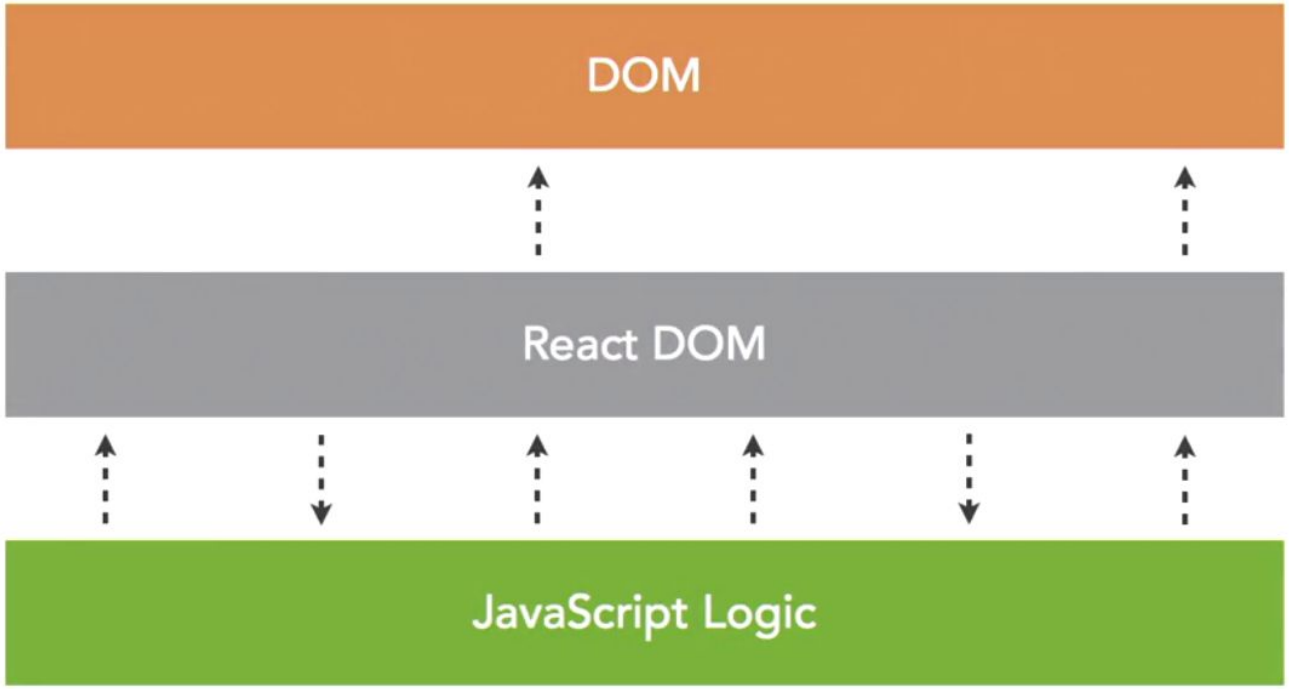
DOM diffing

- Compare la UI actuel avec la nouvelle
- Effectue uniquement les modification nécessaires
- Une comparaison des objets javascript
- Plus rapide que les modification directes sur le DOM

Sans ReactJS



Avec ReactJS (Virtual DOM)



ReactJS: Example

```
1  const app=React.createElement('h1',
2      {id: 'titre', className:'header'},
3      'Bonjour!!');
4
5  ReactDOM.render(app,
6      document.getElementById('root'));
7
```

```
3  <head>
4      <meta charset="utf-8">
5      <title>Ma premiere React app</title>
6      <script type="text/javascript" src='../react.js'></script>
7      <script type="text/javascript" src='../react-dom.js'></script>
8  </head>
9  <body>
10 <div id='root'></div>
11 <script type="text/javascript" src='index.js'></script>
12 </body>
13 </html>
```

ReactJS: Example

```javascript
const {createElement} = React;
const {render} = ReactDOM;

const app=createElement('h1',
    {id: 'titre', className:'header'},
    'Bonjour!!');
render(app, document.getElementById('root'));
```

```html
<head>
    <meta charset="utf-8">
    <title>Ma premiere React app</title>
    <script type="text/javascript" src='../react.js'></script>
    <script type="text/javascript" src='../react-dom.js'></script>
</head>
<body>
<div id='root'></div>
<script type="text/javascript" src='index.js'></script>
</body>
</html>
```

ReactJS: Example

```
1   const {createElement} = React;
2   const {render} = ReactDOM;
3
4   const stl={
5       backgroundColor: 'orange',
6       color: 'white',
7       fontFamily: 'Verdana'
8   };
9
10  const app=createElement('h1',
11      {id: 'titre', className:'header', style: stl},
12      'Bonjour!!');
13  render(app, document.getElementById('root'));
```

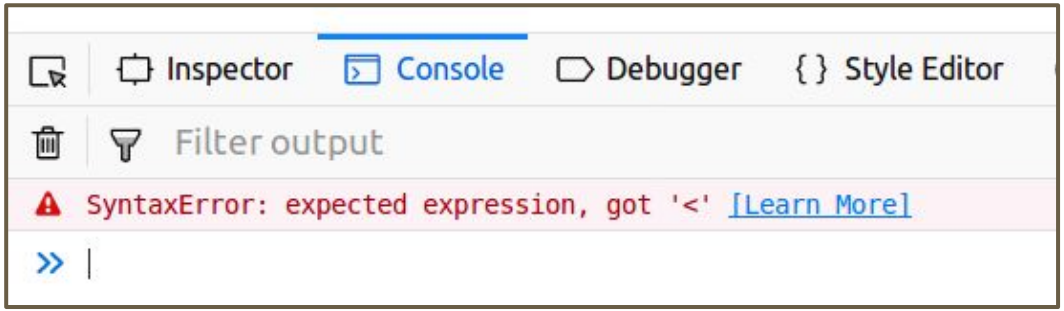
```
title>
src='../react.js'></script>
src='../react-dom.js'></script>
8   </head>
9   <body>
10  <div id='root'></div>
11  <script type="text/javascript" src='index.js'></script>
12  </body>
13  </html>
```

JSX

- Du “***HTML***” dans ***javascript***

```
1  const {render} = ReactDOM;
2
3  const elmt = <h1> Bonjour!! </h1>;
4  render(elmt , document.getElementById('root'));
5
```

⇒ ***Erreur***



Besoin d’un interpréteur ⇒ ***Babel***

JSX: Babel

https://babeljs.io/repl#?babili=false&browsers=&build=&builtIns=false&spec=false&loose=

Docs Setup Try it out Videos Blog Search Donate Team GitHub

```
1 const elmt = <h1> Bonjour!! </h1>;
2 render(elmt , document.getElementById('root'));
```

```
1 'use strict';
2
3 var elmt = React.createElement(
4   'h1',
5   null,
6   ' Bonjour!! '
7 );
8 render(elmt, document.getElementById('root'));
```

JSX: Babel

index.js

```
1  const {render} = ReactDOM;
2
3  const elmt = <h1> Bonjour!! </h1>;
4  render(elmt , document.getElementById('root'));
5
```

index.html

```
3   <head>
4       <meta charset="utf-8">
5       <title>First React app</title>
6       <script type="text/javascript" src='../react.js'></script>
7       <script type="text/javascript" src='../react-dom.js'></script>
8       <script type="text/javascript" src='
        https://cdnjs.cloudflare.com/ajax/libs/
        babel-core/5.8.38/browser.js'></script>
9   </head>
10  <body>
11  <div id='root'></div>
12  <script type="text/babel" src='index.js'></script>
13  </body>
14  </html>
15
```

JSX: Babel

```
1   const {render} = ReactDOM;
2
3   const stl={
4       backgroundColor: 'orange',
5       color: 'white',
6       fontFamily: 'Verdana'
7   };
8
9   const elmt = <h1 id='titre'
10                   className='header'
11                   style={stl}>
12                   Bonjour!!
13              </h1>;
14
15  render(elmt , document.getElementById('root'));
16
```

index.js

```
t="utf-8">
: React app</title>
="text/javascript" src='../react.js'></script>
="text/javascript" src='../react-dom.js'></script>
="text/javascript" src='
s.cloudflare.com/ajax/libs/
.8.38/browser.js'></script>
</head>
<body>
<div id='root'></div>
<script type="text/babel" src='index.js'></script>
</body>
</html>
```

index.html

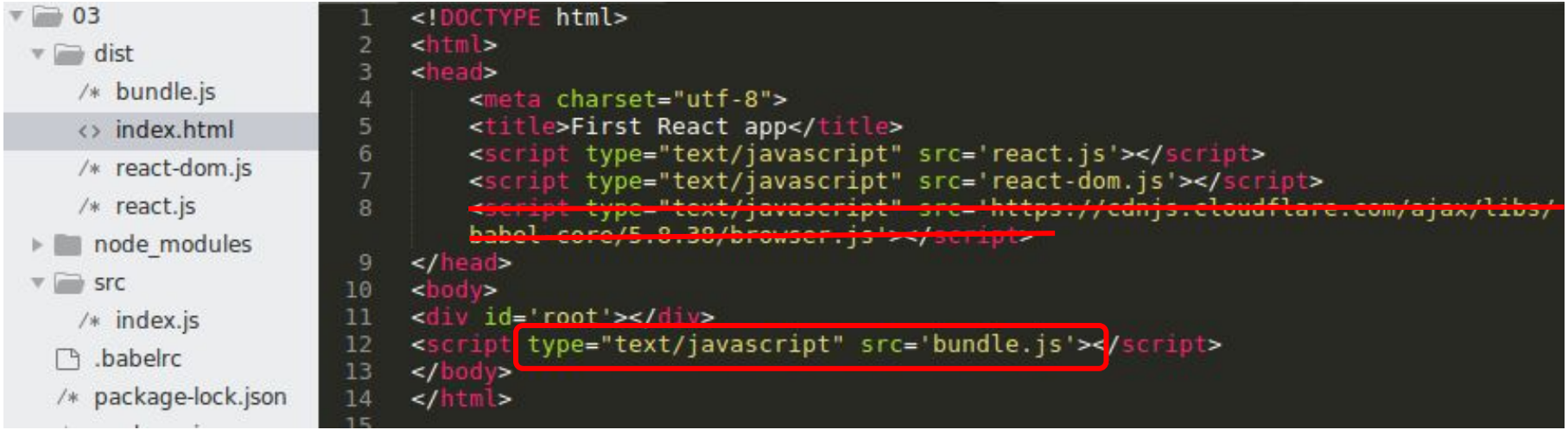
JSX: Babel

- Avec cette méthode, la transformation du jsx en javascript se fait dans le browser.
- Methode valable pour le teste
- Pour la production il faut utiliser le fichier js préalablement transformé

Babel: installation

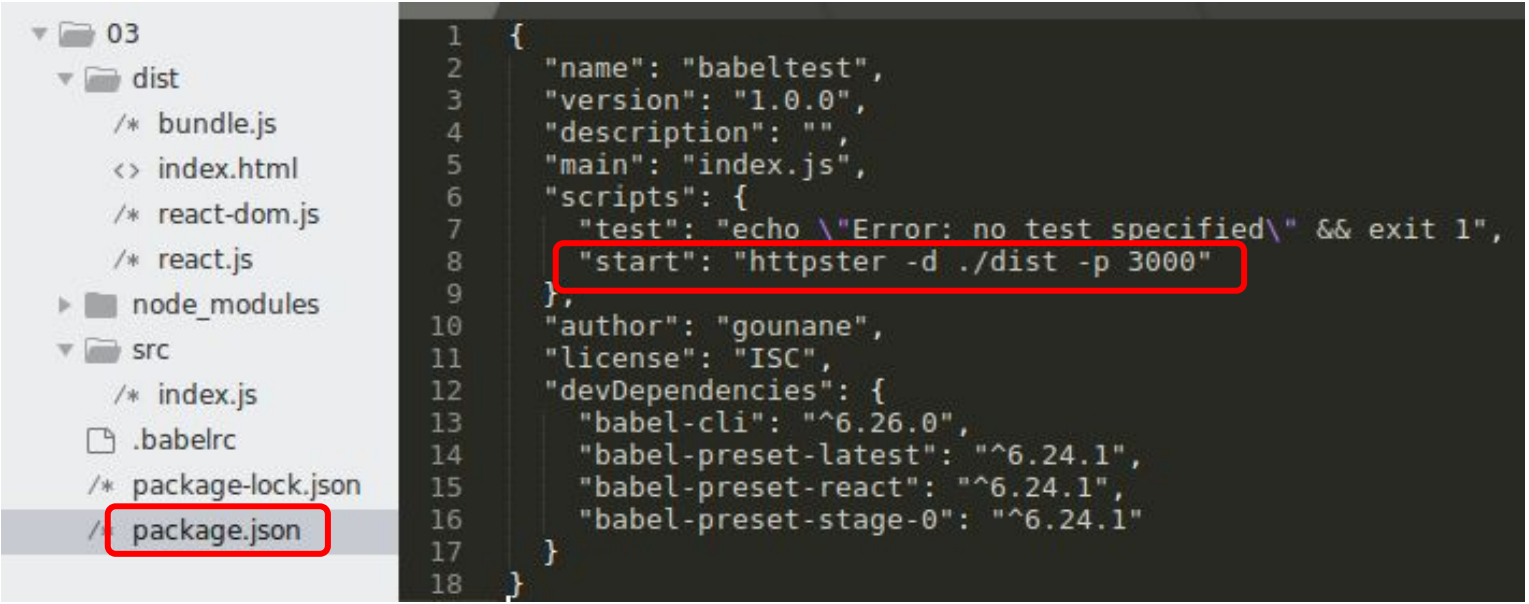
- $ ***npm init*** ⇒ package.json
- $***npm install --save-dev babel-cli*** ou ***sudo npm install -g babel-cli***
- $***npm install --save-dev babel-preset-latest babel-preset-react babel-preset-stage-0***
- Dans le fichier ***.babelrc***: ***{ 'presets': ['latest', 'react', 'stage-0'] }***

JSX: Babel



- ***$./node_modules/.bin/babel ./src/index.js --out-file ./dist/bundle.js*** ou
- ***$babel ./src/index.js --out-file ./dist/bundle.js***

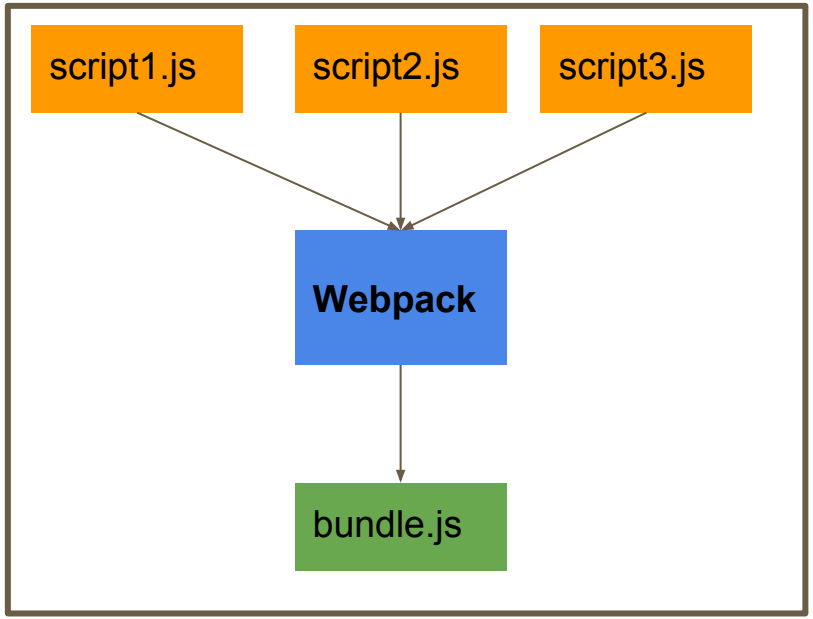
JSX: Babel



- $*npm start*

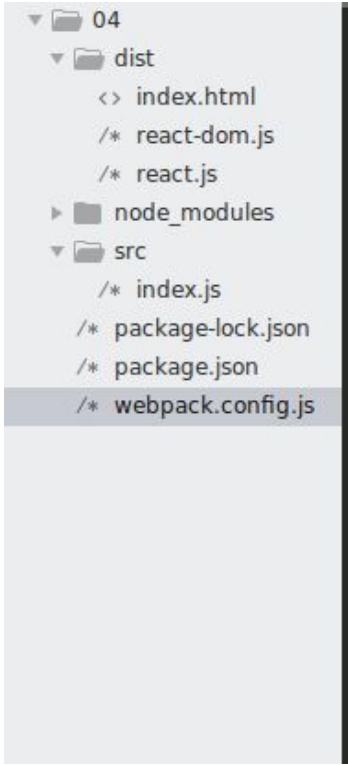
Webpack

- Est un module bundler
- Création des fichiers statiques
- Automatisation des taches



Webpack

- Creer un fichier webpack.config.js




```
1  var webpack = require("webpack");
2
3  module.exports = {
4      entry: "./src/index.js",
5      output: {
6          path: "dist/assets",
7          filename: "bundle.js",
8          publicPath: "assets"
9      },
10     devServer: {
11         inline: true,
12         contentBase: './dist',
13         port: 3000
14     },
15     module: {
16         loaders: [
17             {
18                 test: /\.js$/,
19                 exclude: /(node_modules)/,
20                 loader: ["babel-loader"],
21                 query: {
22                     presets: ["latest", "stage-0", "react"]
23                 }
24             }
25         ]
26     }
27 }
```

Webpack

- $ ***npm install webpack babel-loader webpack-dev-server --save-dev***
- $***./node_modules/.bin/webpack***
- Dans le fichier ***package.json*** modifier la ligne “start”:
  - ***“start”: “./node_modules/.bin/webpack-dev-server”***
- ***$ npm start***

```
5       <title>First React app</title>
6       <script type="text/javascript" src='react.js'></script>
7       <script type="text/javascript" src='react-dom.js'></script>
8   </head>
9   <body>
10  <div id='root'></div>
11  <script type="text/javascript" src='assets/bundle.js'></script>
12  </body>
```

Webpack

FOLDERS

- 04
  - dist
    - assets
      - bundle.js
    - index.html
    - react-dom.js
    - react.js
  - node_modules
  - src
    - index.js
  - package-lock.json
  - package.json
  - webpack.config.js

index.html

```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <title>First React app</title>
    <script type="text/javascript" src='react.js'></script>
    <script type="text/javascript" src='react-dom.js'></script>
</head>
<body>
<div id='root'></div>
<script type="text/javascript" src='assets/bundle.js'></script>
</body>
</html>
```

Webpack

- *$ npm install --save react react-dom*
- *$./node_modules/.bin/webpack*
- *$ npm start*

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <meta charset="utf-8">
5      <title>First React app</title>
6      <script type="text/javascript" src='react.js'></script>
7      <script type="text/javascript" src='react-dom.js'></script>
8  </head>
9  <body>
10 <div id='root'></div>
11 <script type="text/javascript" src='assets/bundle.js'></script>
12 </body>
13 </html>
```

```
1  const {render} = ReactDOM;
2  import React from "react";
3  import ReactDOM from "react-dom";
4  const stl={
5      backgroundColor: 'orange',
6      color: 'white',
7      fontFamily: 'Verdana'
8  };
9
10 const elmt = <h1 id='titre'
11                 className='header'
12                 style={stl}>
13                 Bonjour!!
14             </h1>;
15
16 ReactDOM.render(elmt , document.getElementById('root'));
17
```

Webpack

Avec webpack on peut importer des fichiers et des modules:

Creer une application react sans configuration

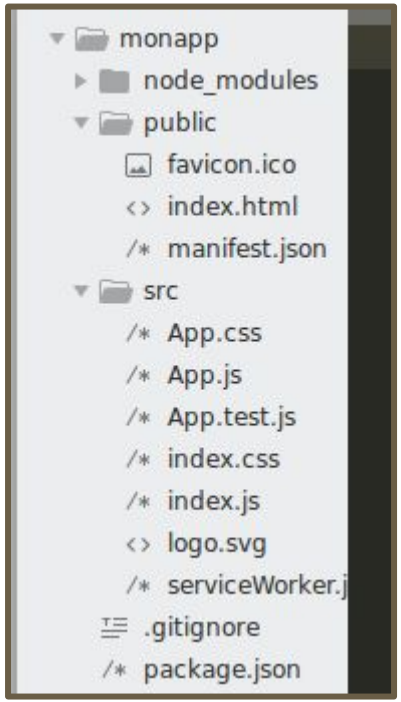
Pour créer une application reactjs sans passer par les étapes précédentes, la commande create-react-app se charge de toutes les configurations nécessaires.

Pour installer create-react-app:

$***sudo npm install -g create-react-app***

Pour créer une application “***monapp***”:

$ ***create-react-app monapp***



Lancer une application react

$ ***cd monapp***

$ ***npm start***

Principe

Index.html

```html
  <body>
    <noscript>
      You need to enable JavaScript to run this app.
    </noscript>
    <div id="root"> </div>
  </body>
</html>
```

index.js

```javascript
import React from 'react';
import ReactDOM from 'react-dom';
import './index.css';
import App from './App';

ReactDOM.render(<App />, document.getElementById('root'));
```

React Component

1- Étendre la classe Component

2- Redefinir render()

3- JSX

4- export pour l'utiliser dans un autre fichier js

```
1  import React, { Component } from 'react';
2  import logo from './logo.svg';
3  import './App.css';
4
5  class App extends Component {
6    render() {
7      const titre="MonApp React";
8      return (
9        <div className="App">
10           <h1>{titre}</h1>
11           <p>Voice ma premiere application ReactJS</p>
12       </div>
13     );
14   }
15 }
16
17 export default App;
```

Les Props

Pour passer des arguments à un ***Component*** on utilise les ***props***

<Moncomponent arg1="val1" arg2="val2" arg3={js expression} />

Pour accéder à ces props on a accès à l'objet props dans la méthode render de moncomponent.js

- ***this.props.arg1***
- ***this.props.arg2***
- ***this.props.arg3***

Les props

app.js

```
5  class App extends Component {
6    render() {
7      const titre=this.props.titre;
8      return (
9        <div className="App">
10             <h1>{titre}</h1>
11             <p>Une premiere application ReactJS par {this.props.nom}</p>
12       </div>
13     );
14   }
15 }
```

index.js

```
1  import React from 'react';
2  import ReactDOM from 'react-dom';
3  import './index.css';
4  import App from './App';
5
6  ReactDOM.render(<App titre="ReactJS" nom="Yasser" />, document.getElementById('root'));
7
```

Le state

1. L'objet ***state*** contient des données utilisées par le ***Component***
2. Les changement dans cet objet seront répercutés automatiquement sur ce ***component***.
3. La modification de l'objet state ne se fait que par la méthode ***setStat(newstate)***;

```
1  import React, { Component } from 'react';
2  import './style/bootstrap.min.css';
3  import './style/App.css';
4  import {textVal} from "./textVal";
5
6  class App extends Component {
7    state={
8      text: textVal
9    }
10   render() {
11     const titre=this.props.titre;
12     return (
13       <div className="container">
14             <div className="titre">
15               <h1>{titre}</h1>
16               <p>Par: {this.props.nom}</p>
17             </div>
18             <div className="row">
19               <div className="col-sm-6">
20                 <textarea rows="20" className="form-control"
21                 value={this.state.text}></textarea>
22               </div>
23               <div className="col-sm-6">
24                 <h1>Resultats</h1>
25               </div>
26             </div>
27       </div>
28     );
29   }
30 }
```

Le state

1. Lors de l'édition dans le textArea
2. Les modification sont automatiquement apporté au div

```
6   class App extends Component {
7     state={
8       text: ""
9     }
10    handleEdite=(e)=>{
11      const t=e.target.value;
12      const newState={
13        text: t
14      };
15      this.setState(newState);
16    }
17    render() {
18      const titre=this.props.titre;
19      return (
20        <div className="container">
21          <div className="titre">
22            <h1>{titre}</h1>
23            <p>Par: {this.props.nom}</p>
24          </div>
25          <div className="row">
26            <div className="col-sm-6">
27              <textarea rows="20" className="form-control"
28              onChange={(e)=> this.handleEdite(e)}></textarea>
29            </div>
30            <div className="col-sm-6">
31              <h1>{this.state.text}</h1>
```

Le cycle de vie d’un Component

- Mounting
- Updating
- Unmounting

Le cycle de vie d’un Component

- Mounting
  - constructor()
  - componentWilmount()
  - render()
  - ComponentDidMount()

Le cycle de vie d’un Component

- Updating
  - getDerivedStateFromProps()
  - shouldComponentUpdate()
  - render()
  - getSnapshotBeforeUpdate()
  - componentDidUpdate()

Le cycle de vie d’un Component

- Unmounting
  - componentWillUnmount()

Exercice

Chercher Un Utilisateur
search
said gounane
sg@yahoo.com
Ayman baroudi
ib@gmail.com
Read Aftati
reda@yahoo.com